

Position



Categories

- ❑ **Absolute:** Send out the correct position right after they are powered. No movement to a reference position is required.
- ❑ **Relative:** Count pulses from a reference position. A movement to a reference (zero) position is required.

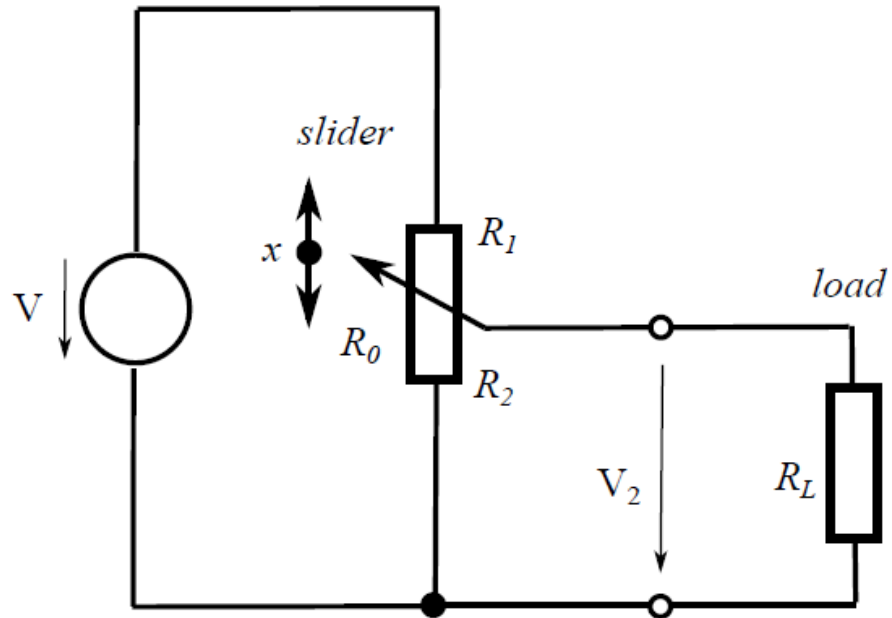
Categories

- ❑ **Continuous** - they measure continuously the distance between the sensor and the object. The output signal is proportional to the measured distance.
- ❑ **Proximity** - proximity sensors detect only the presence of the object or not its distance. The output is a digital signal, 0/1, object is present or not.

Categories

- ❑ **Linear** - the sensor measures in principle linear displacement
- ❑ **Angular** - the sensor measures in principle angular displacement

Resistive sensor



Resistive sensor

$$R_2 = xR_0$$

$$R_1 = R_0 - R_2 = R_0 - xR_0 = (1 - x) R_0$$

$$k = \frac{R_L}{R_0}$$

$$\begin{aligned} V_2 &= V \frac{x \cdot k \cdot R_0 \cdot R_0}{(1-x)R_0 \cdot x \cdot R_0 + (1-x)R_0 \cdot k \cdot R_0 + x \cdot k \cdot R_0 R_0} = \\ &= V \frac{x \cdot k \cdot R_0^2}{x \cdot R_0^2 - x^2 \cdot R_0^2 + x \cdot R_0^2 - x \cdot k \cdot R_0^2 + x \cdot k \cdot R_0^2} = \\ &= V \frac{k \cdot x}{(1-x)x + k} \end{aligned}$$

Resistive sensor

□ Relative error of resistive position sensor

$$\begin{aligned}\delta [\%] &= \frac{V_2 - x \cdot V}{V} \cdot 100 = \frac{V \frac{k \cdot x}{(1-x)x+k} - x \cdot V}{V} \cdot 100 = \\ &= -\frac{(1-x)x^2}{(1+x)x+k} \cdot 100\end{aligned}$$

The position of maximal relative error

$$\frac{\partial \delta}{\partial x} = 0 \Rightarrow x = 1/2$$

Resistive sensor

□ Absolute error

$$\Delta[V] = V \frac{k \cdot x}{(1 - x)x + k} - x \cdot V$$

The position of maximal absolute error is dependent on the value of k . For example for $k = 1$ the position of maximal absolute error is

$$\frac{\partial \Delta}{\partial x} = 0 \rightarrow x \approx 0,69$$

Inductive sensors

Measure small distances, up to approximately 1 mm.

The coil inductivity

$$L = \frac{N^2}{R_m}$$

R_m : magnetic resistance (reluctance); N: number of turns

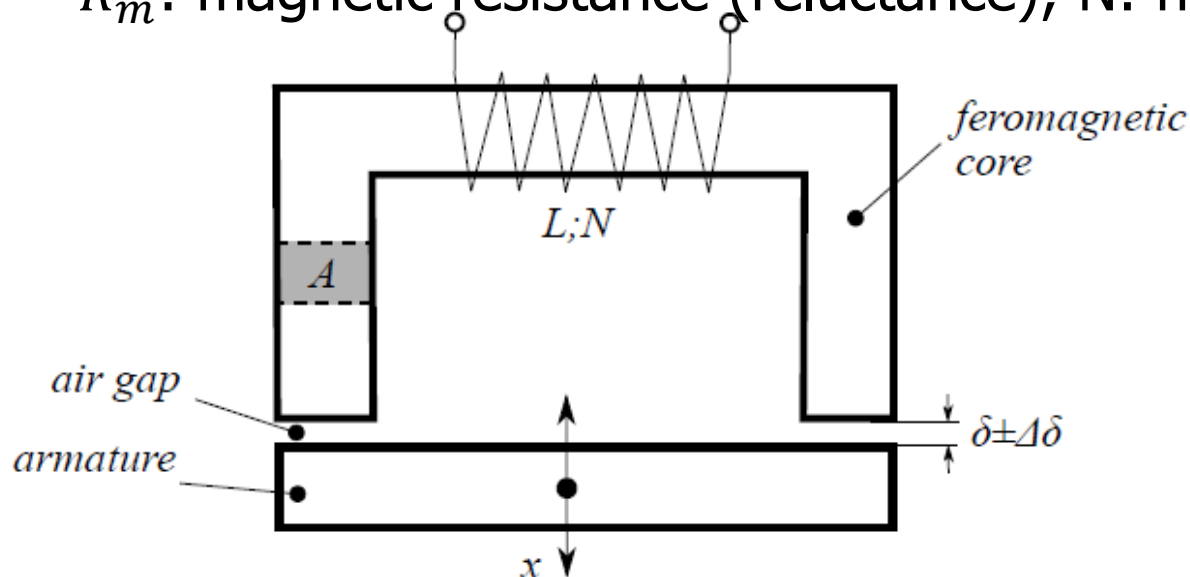


FIGURE 7.8: Small air gap sensor

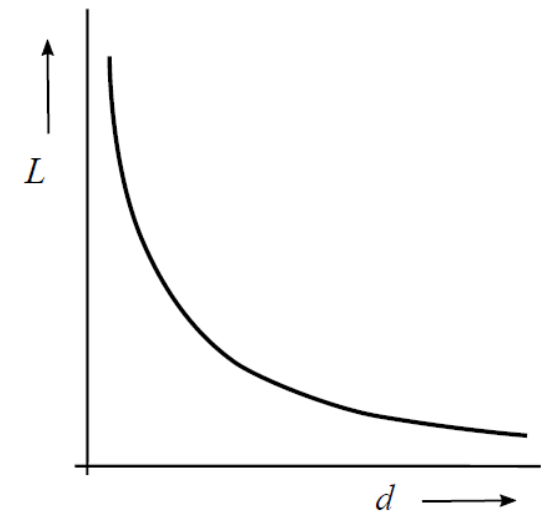


FIGURE 7.9: Inductance as a function of air-gap size

Inductive sensors

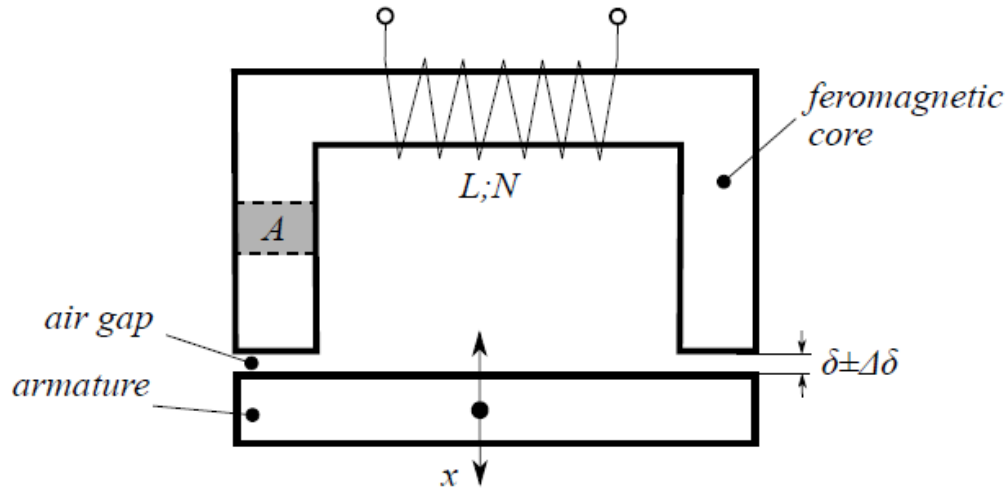


FIGURE 7.8: Small air gap sensor

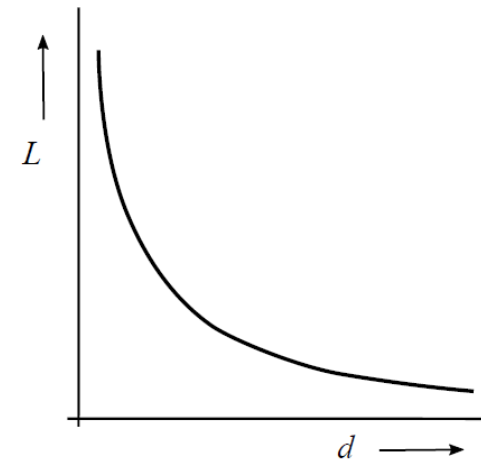


FIGURE 7.9: Inductance as a function of air-gap size

- Reluctance of uniform magnetic circuit: $R = \frac{l}{\mu A}$
- Reluctance = air-gap reluctance R_{ag} + iron reluctance R_{Fe}

$$R_m = R_{Fe} + R_{ag} = \left(\frac{l_{Fe}}{\mu A} \right) + \left(\frac{2\delta}{\mu A} \right)$$

l_{Fe} : length of iron magnetic circuit

μ : the permeability of the material

Inductive sensors

- Assuming $R_{ag} \gg R_{Fe}$

$$L = \frac{\mu AN^2}{2\sigma}$$

- The impedance change causes the change of inductor current.

$$\hat{I} = \frac{\hat{V}}{\hat{Z}} = \frac{\hat{V}}{R + j\omega L}$$

Assuming $\frac{\omega L}{R} \gg 1$

$$I = \frac{V}{\omega L} = \frac{2V}{N^2 \mu_0 A} \times \delta$$

Inductive sensors

- **Region 1:** $R_{ag} \gg R_{Fe}$ is not valid. The dependence is non-linear
- **Region 2:** The dependence is approximately linear.
- **Region 3:** $\omega L/R \gg 1$ is not valid. Increase in the air gap does not change the current.

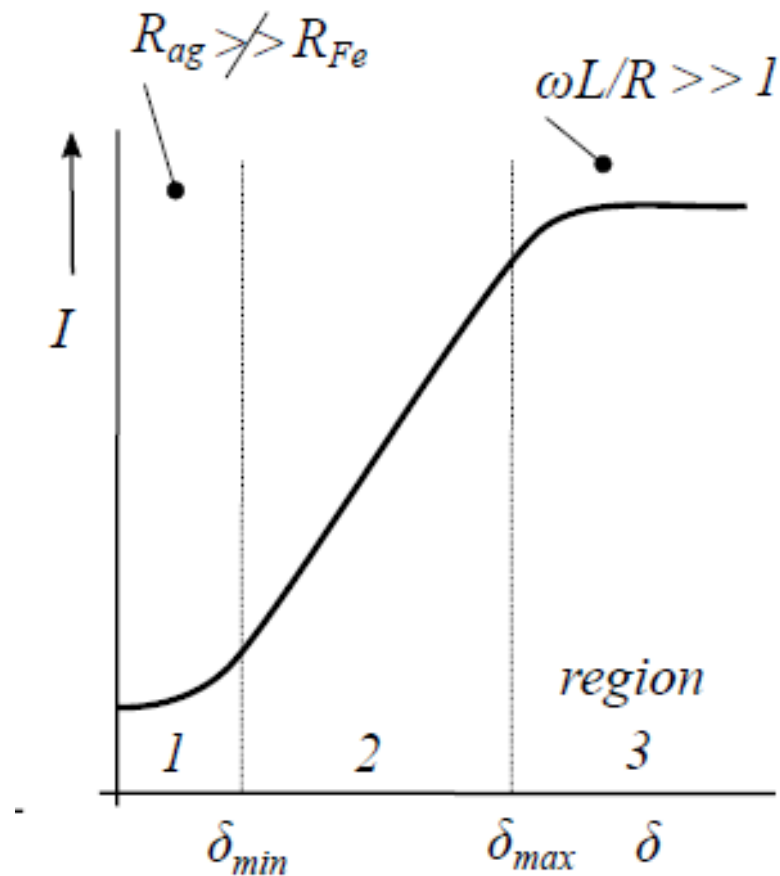
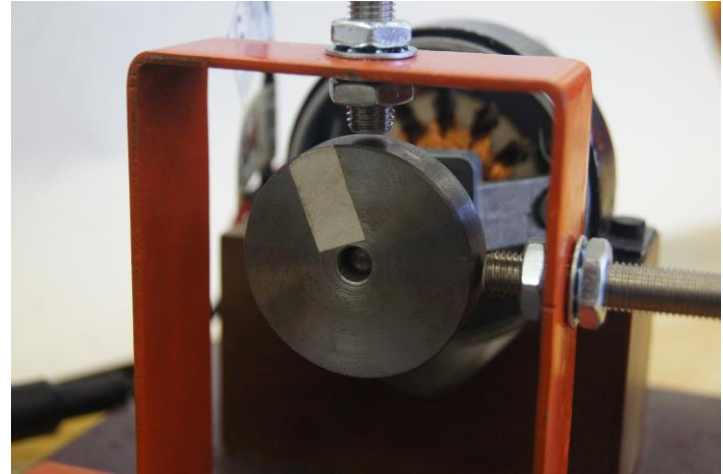
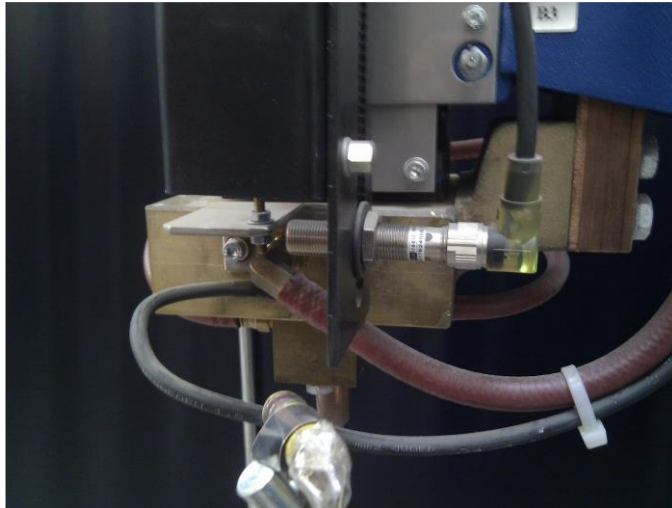


FIGURE 7.10: Inductor current I as a function of air-gap size

Inductive sensors



Linear variable differential transformer (LVDT)

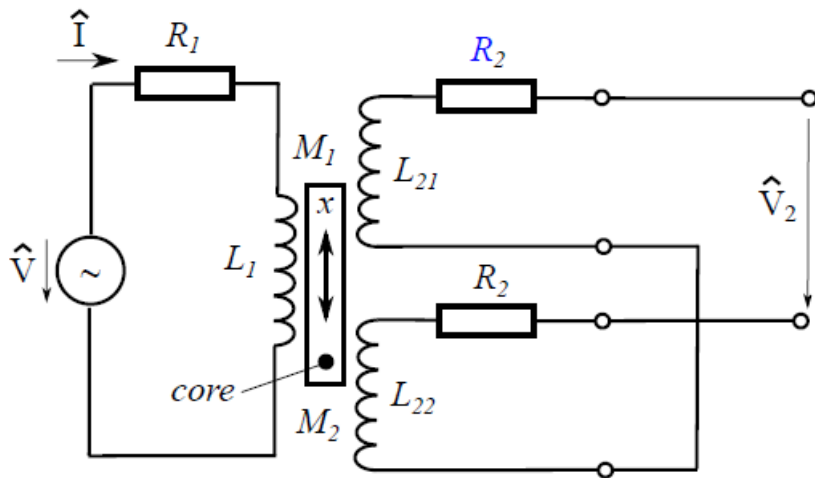


FIGURE 7.14: LVDT

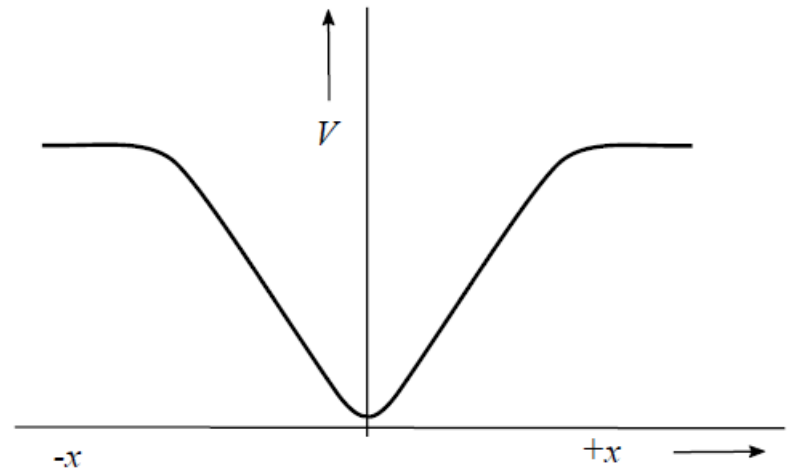
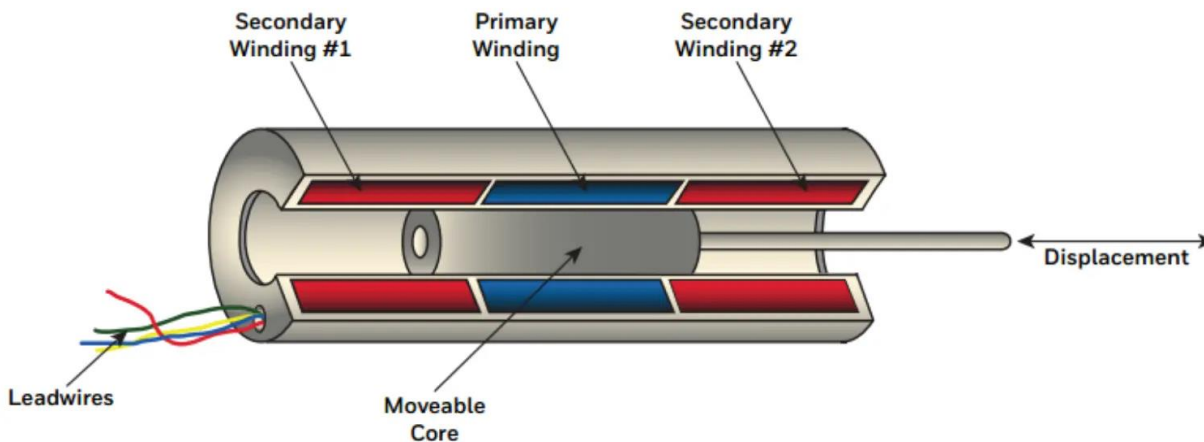


FIGURE 7.15: Steady-state characteristic of LVDT



Resolver

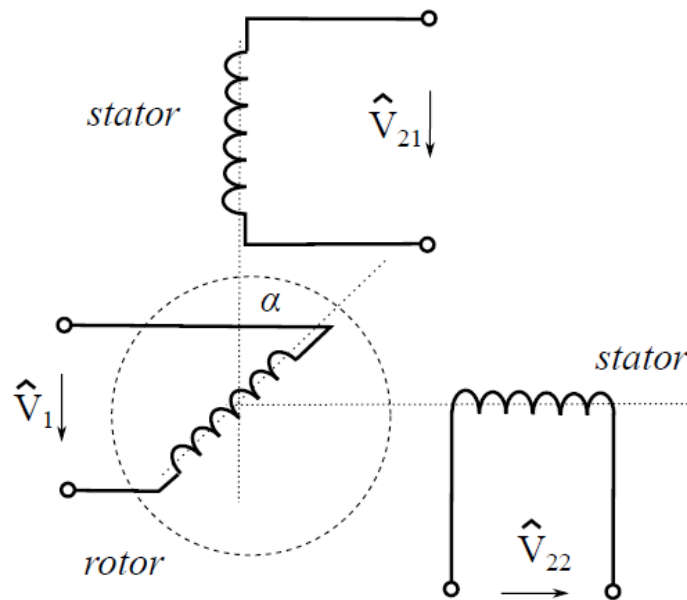


FIGURE 7.18: Principle of resolver



$$\hat{V} = V_{max} \sin \omega t$$

□ Induced voltage in stator winding

$$\hat{V}_{21} = V_{max} \sin \omega t \sin \alpha$$

$$\hat{V}_{22} = V_{max} \sin \omega t \cos \alpha$$

Resolver

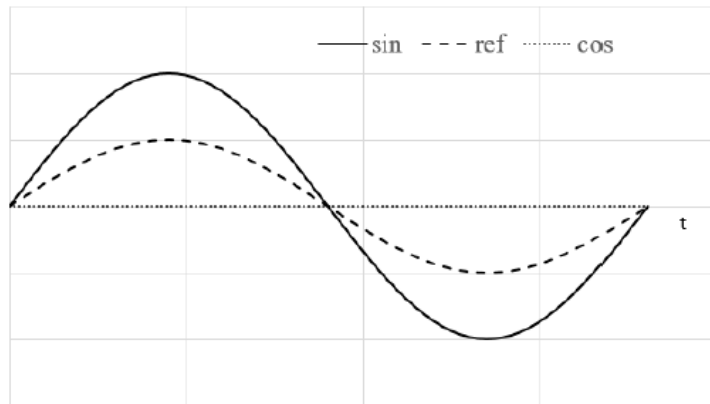


FIGURE 7.21: Resolver signal in position 0° , signal SIN with maximal amplitude, signal COS with amplitude zero

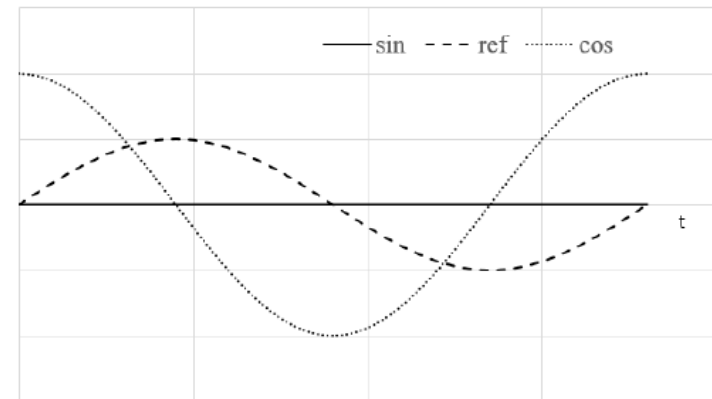
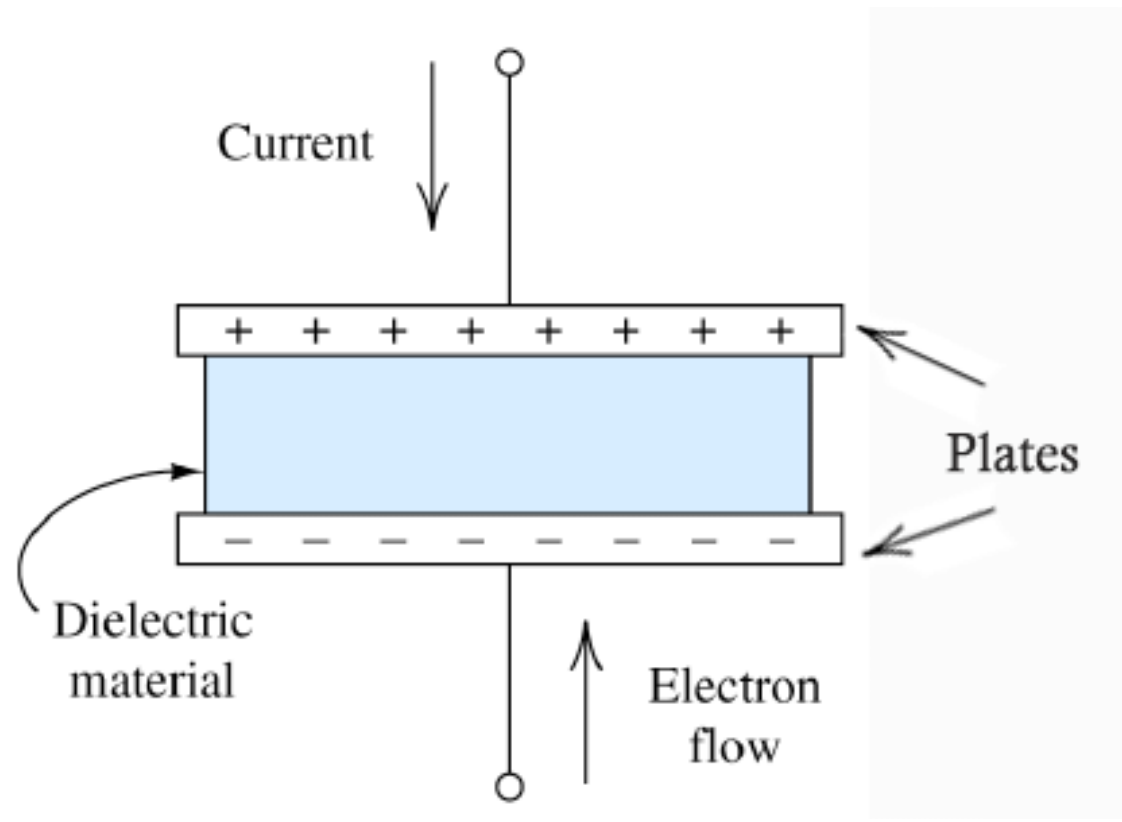


FIGURE 7.22: Resolver signal in position 90° , signal COS with maximal amplitude, signal SIN with amplitude zero

Capacitive sensors

□ For a planar capacitor

$$C = \epsilon_0 \epsilon_r \frac{A}{d}$$



Capacitive sensors

□ Sensors based on change of distance between electrodes

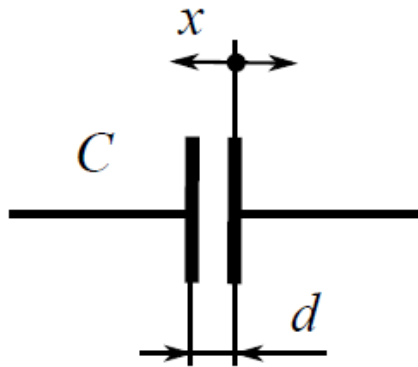


FIGURE 7.24: Capacitive sensor with change of distance between electrodes [6]

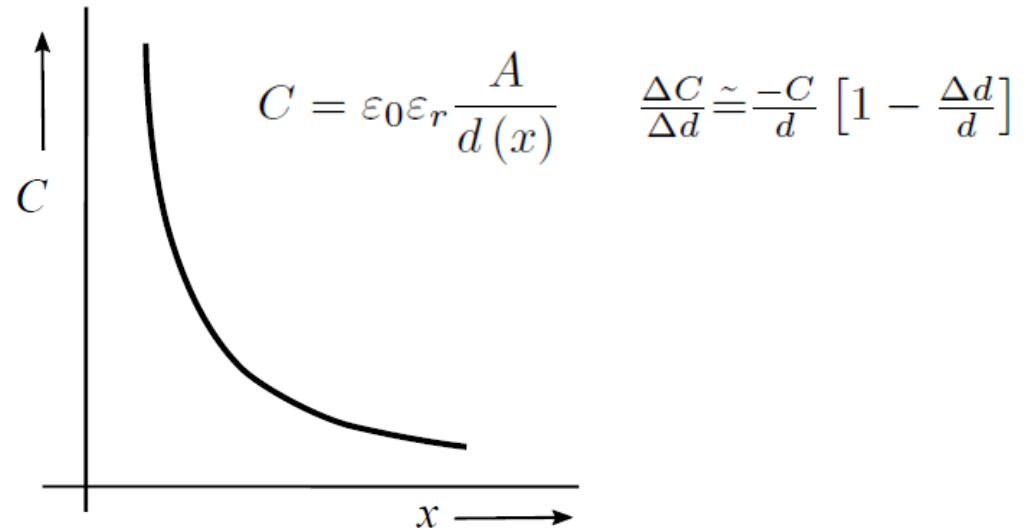


FIGURE 7.25: Steady-state characteristics of a capacitive sensor with change of distance between electrodes [6]

Capacitive sensors

□ Differential capacitor

$$C_1 = \varepsilon_0 \varepsilon_r \frac{A}{d_1(x)}; \quad C_2 = \varepsilon_0 \varepsilon_r \frac{A}{d_2(x)}$$

Sensitivity

$$\frac{\Delta C}{\Delta d} = -\frac{C}{d} \left[1 - 2 \left(\frac{\Delta d}{d} \right)^2 \right]$$

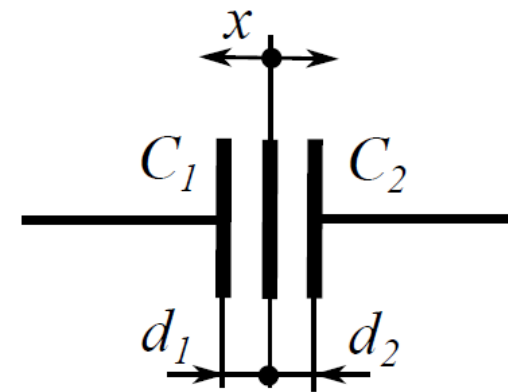
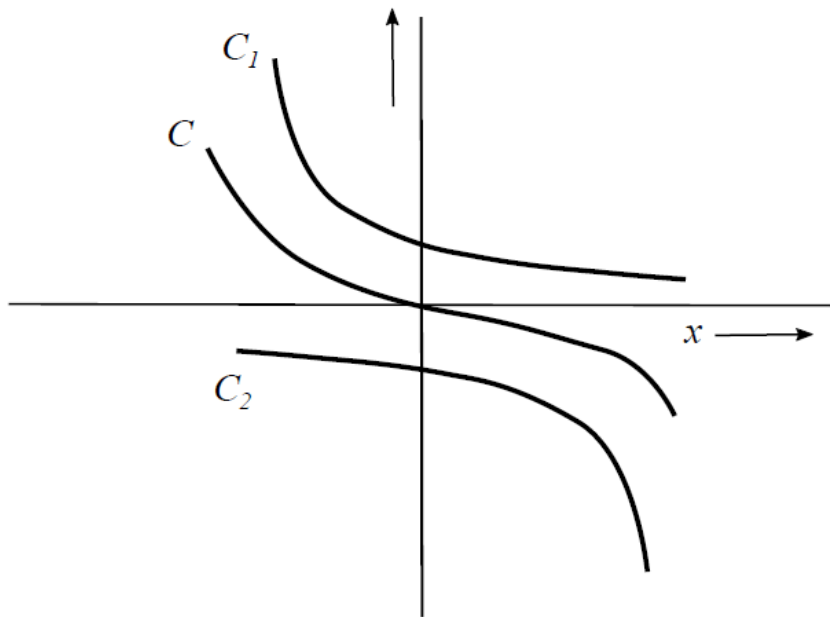


FIGURE 7.27: Steady-state characteristic of a differential capacitive position sensor [6]

FIGURE 7.26: Differential capacitor [6]

Capacitive sensors

□ Differential capacitor

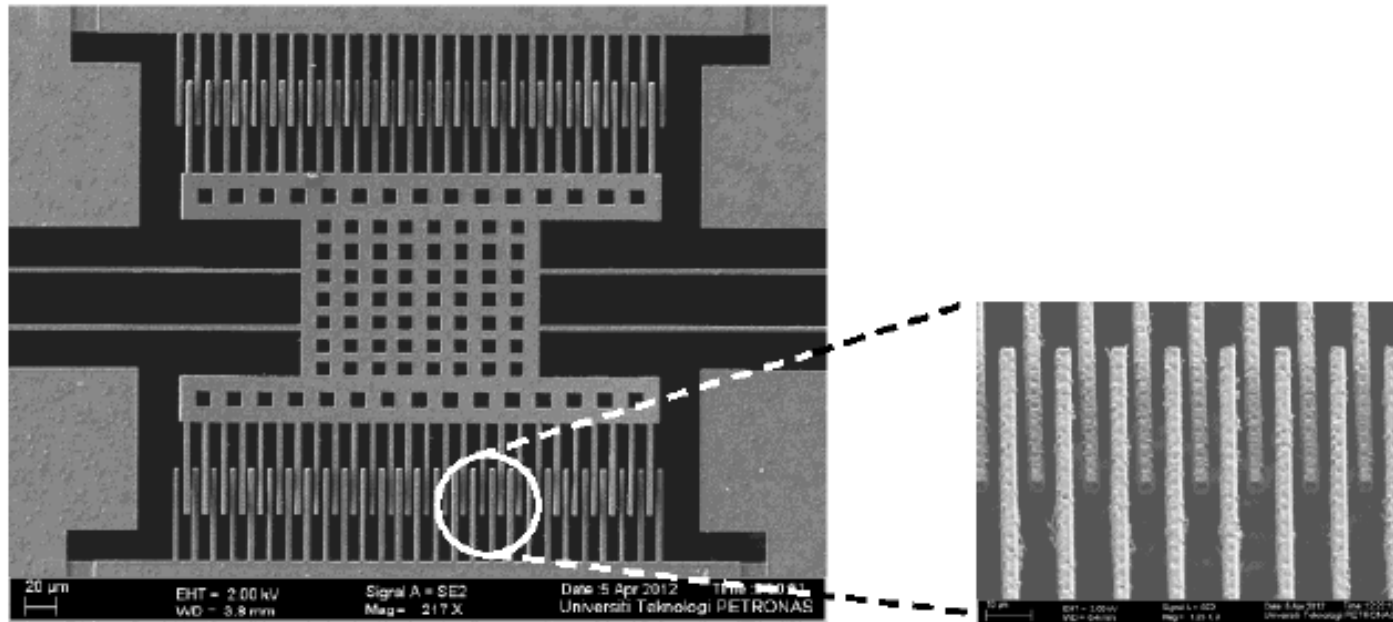


FIGURE 7.28: Example of a differential capacitive sensor in a magnetic field sensor, courtesy of [40], license CC BY 3.0

Capacitive sensors

□ Change of electrode area

$$C = \varepsilon_0 \varepsilon_r \frac{A(x)}{d}$$

Sensitivity [6]

$$\frac{\Delta C}{\Delta l} \cong -\frac{C}{d} \left[1 + \frac{\Delta d}{d} \right]$$

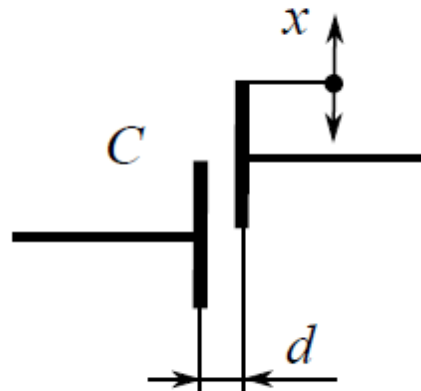


FIGURE 7.29: Capacitive sensor with change of area [6]

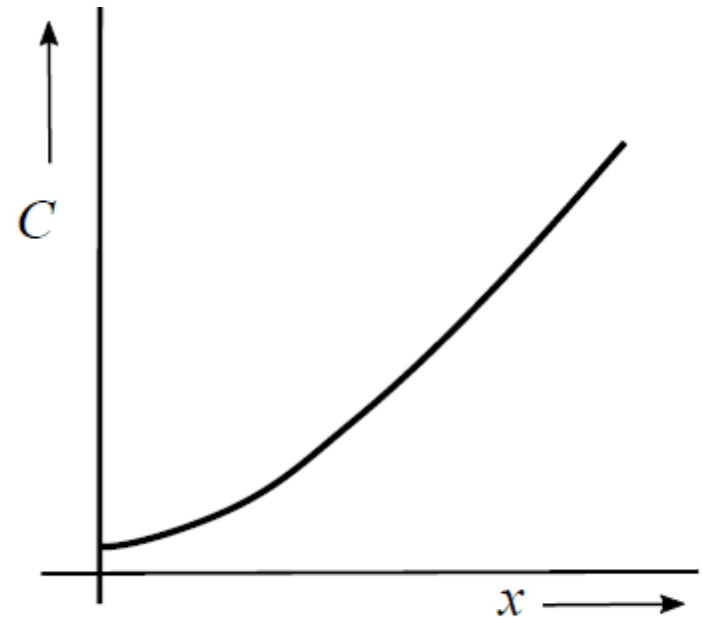


FIGURE 7.30: Steady-state characteristic of a capacitive sensor with area change [6]

Capacitive sensors

□ Change of permittivity

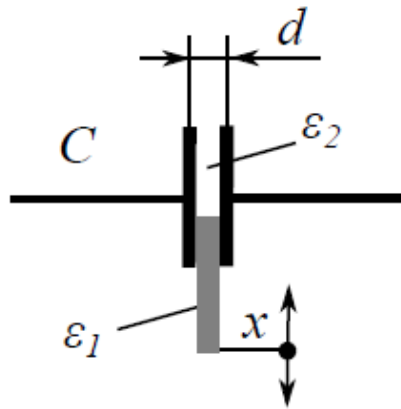


FIGURE 7.31: Capacitive sensor with permittivity change [6]

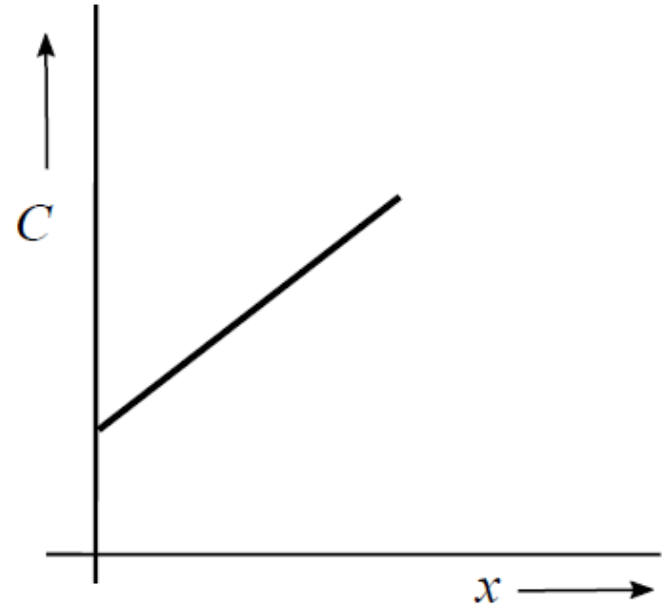
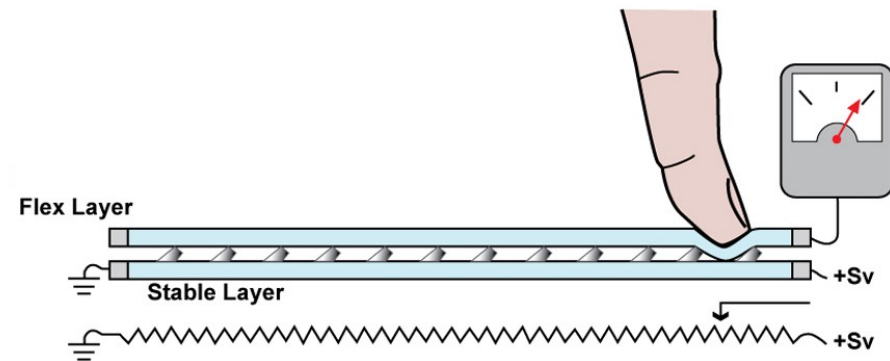
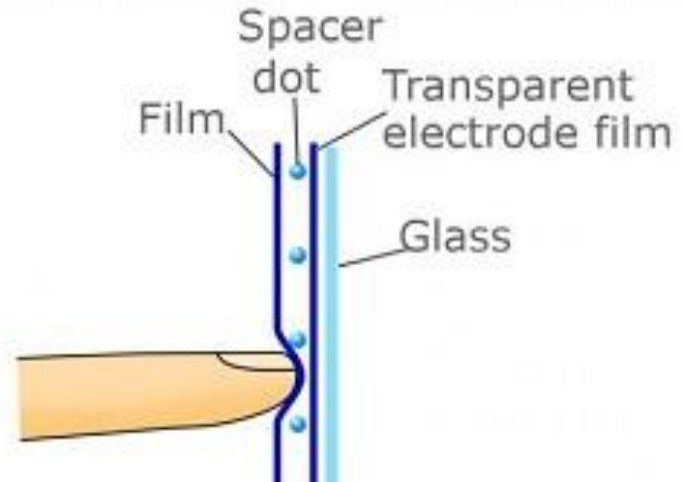
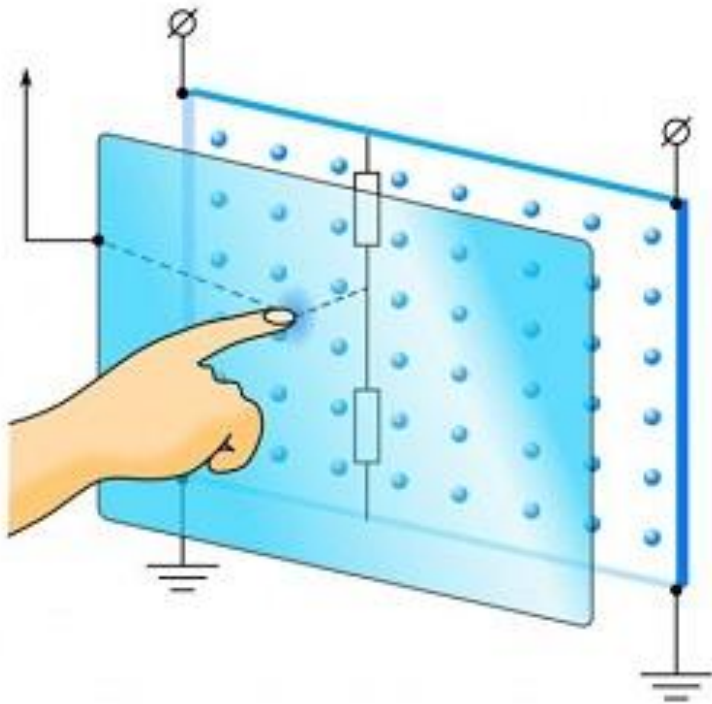


FIGURE 7.32: Steady-state characteristic of a capacitive sensor with permittivity change [6]

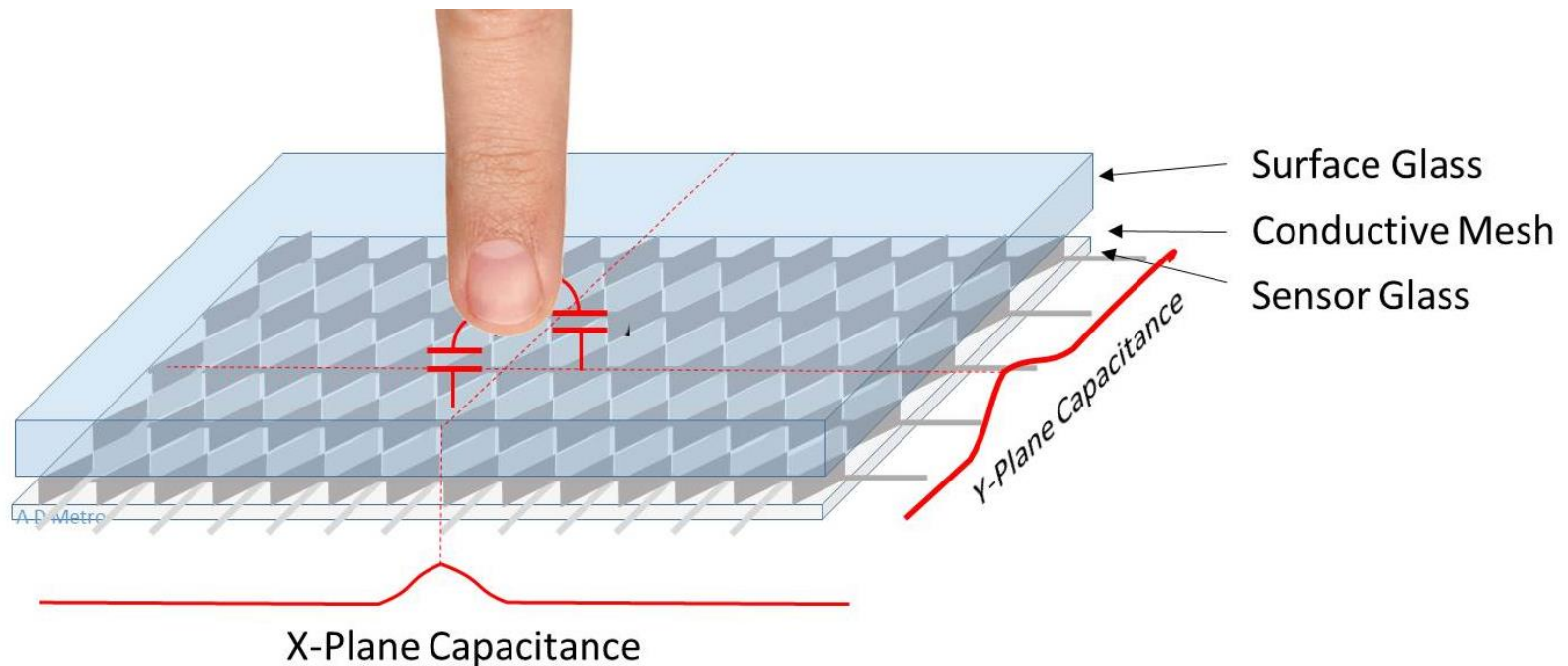
Capacitive sensors

❑ Resistive Touchscreen



Capacitive sensors

□ Capacitive Touchscreen



Capacitive sensors

❑ Optic Fingerprint Reader

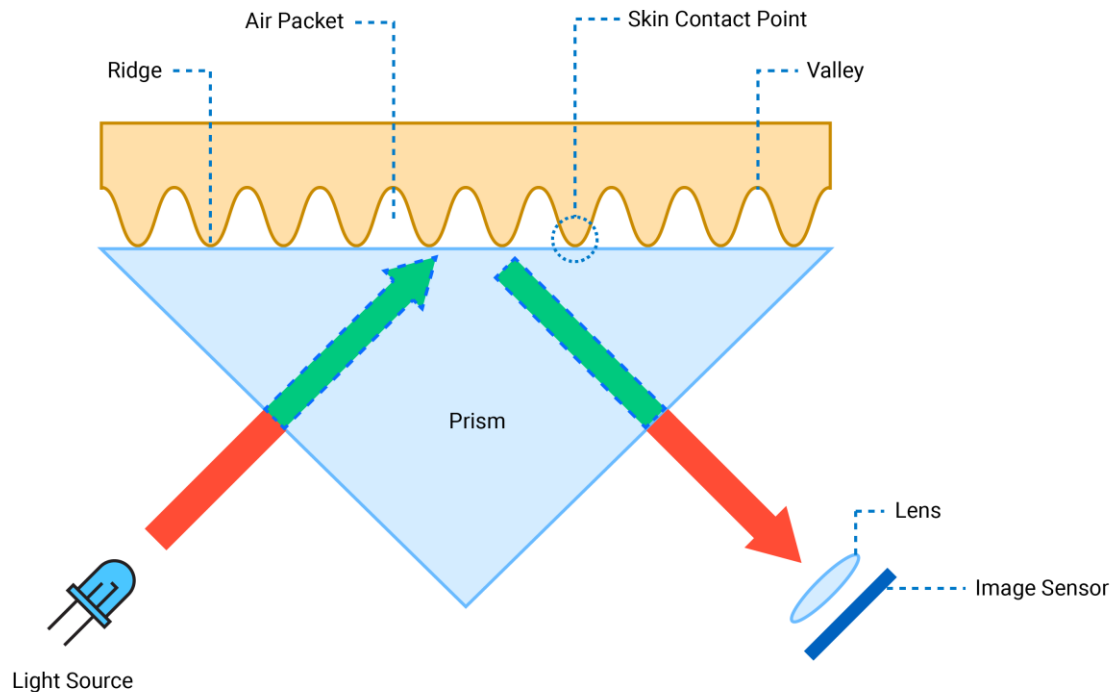
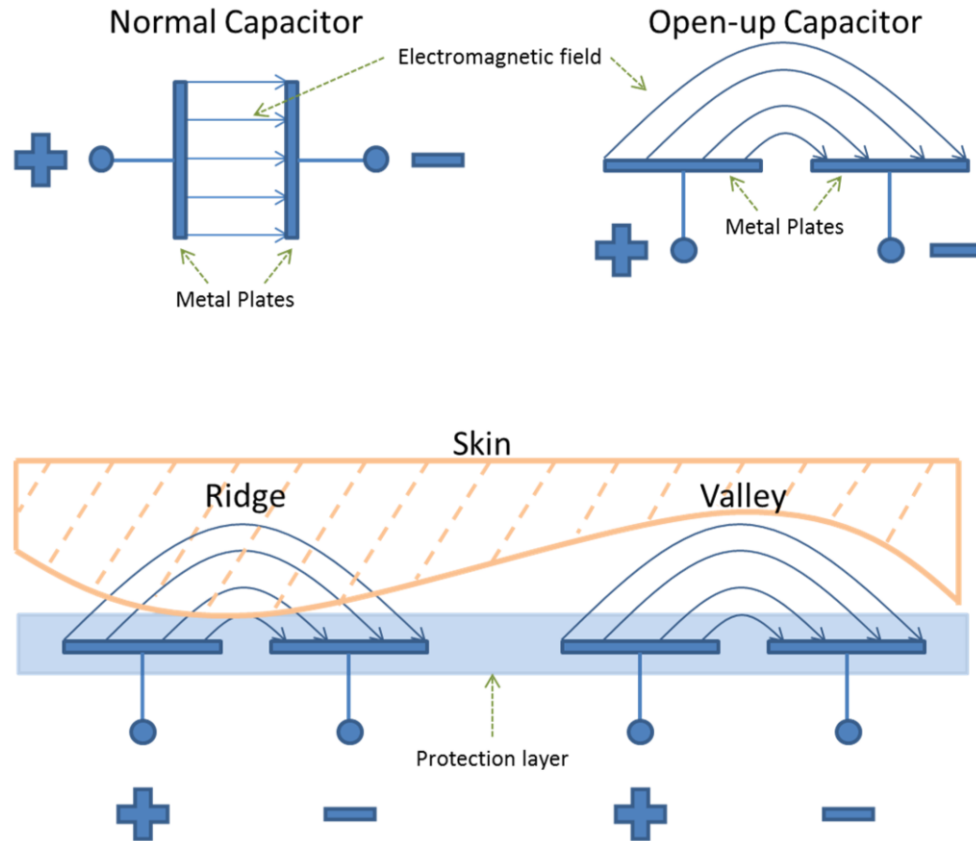


Fig. Optical Fingerprint Scanner Working Principle



Capacitive sensors

□ Capacitive fingerprint sensor

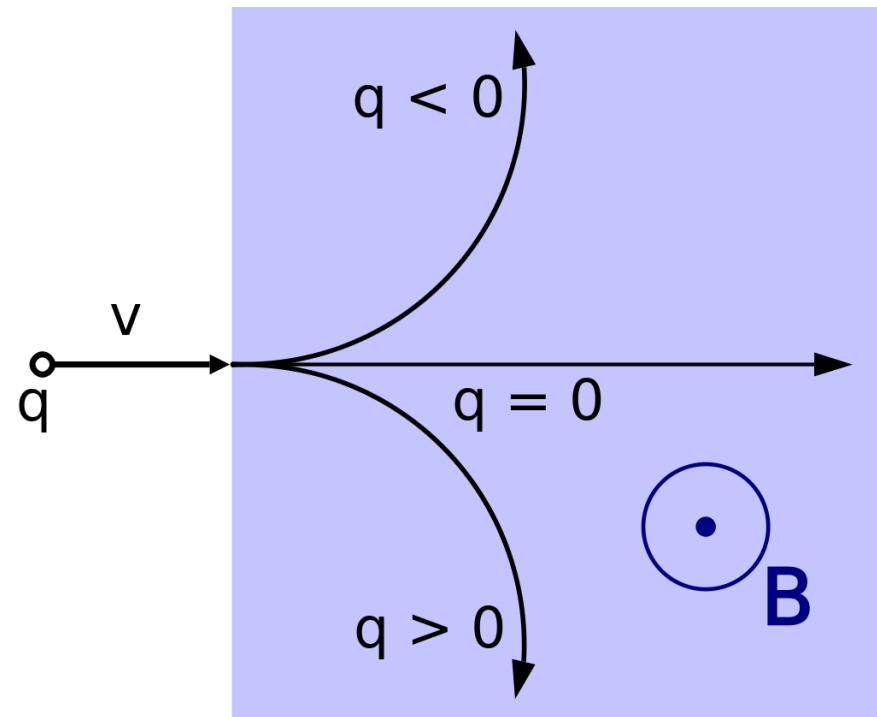
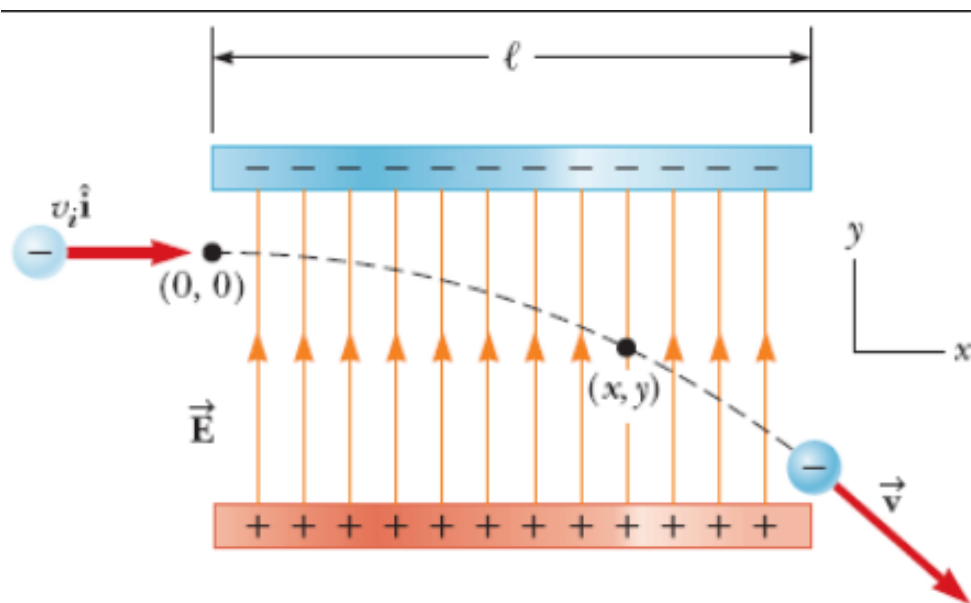


(c) Capacitive fingerprint sensor measurement principle.

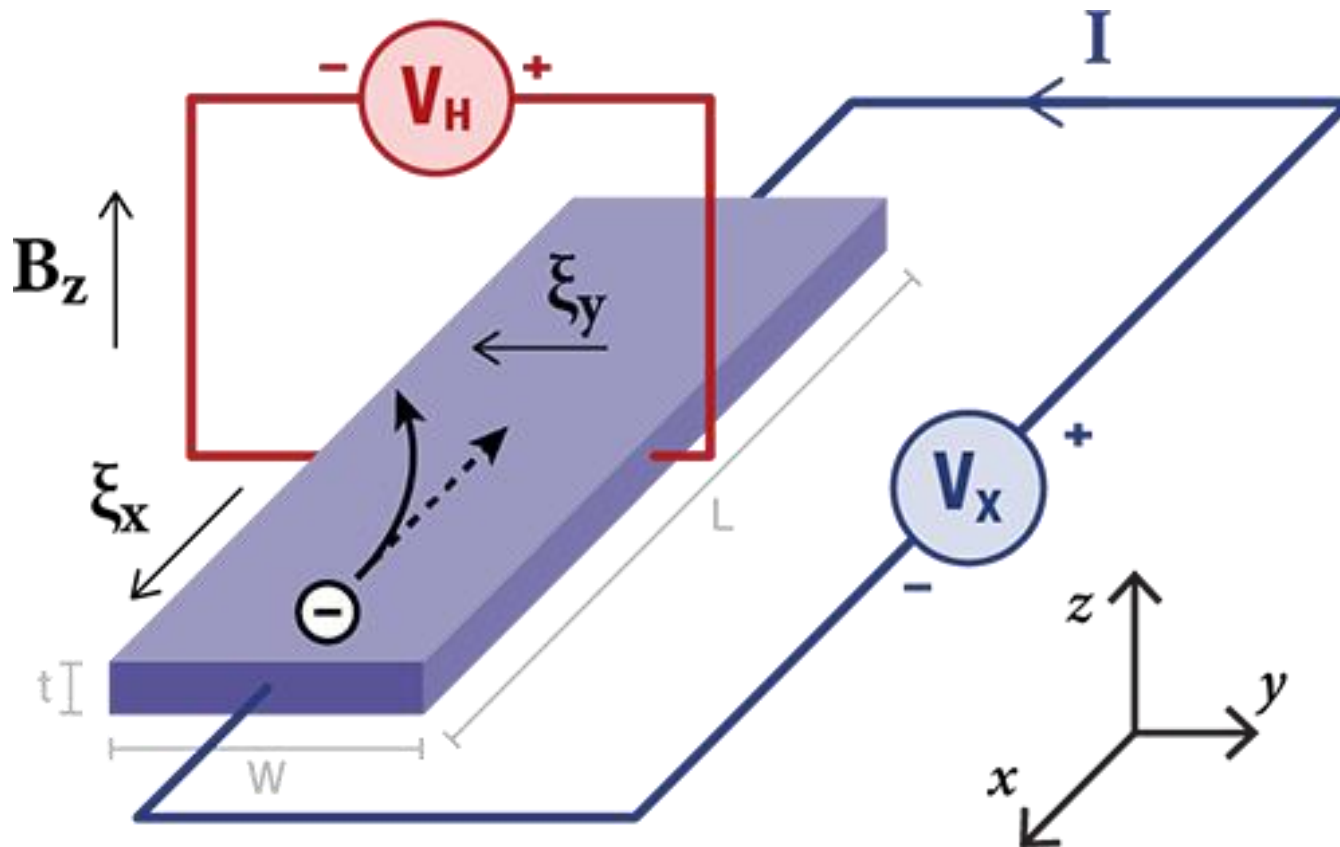
Magnetic (Hall) sensors

Lorentz force

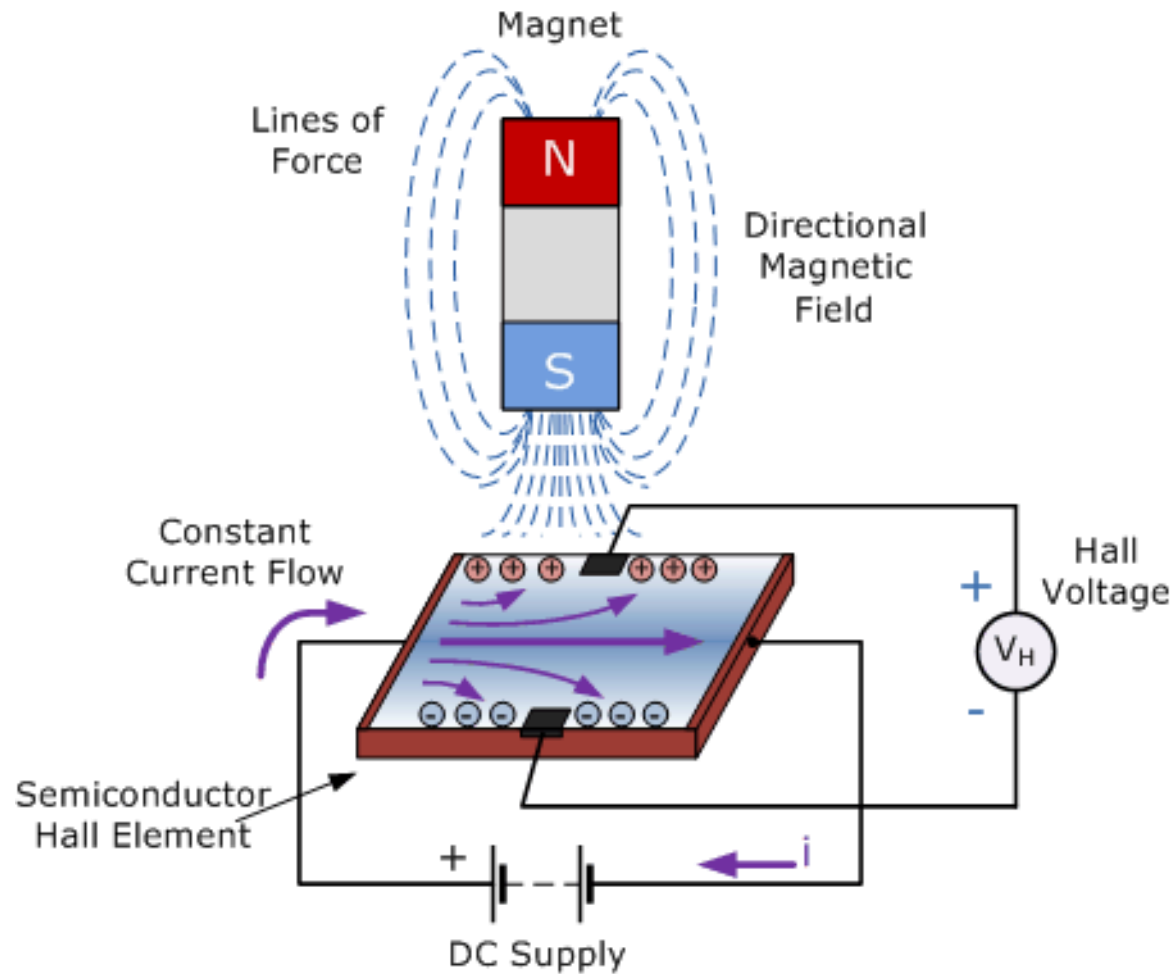
$$\mathbf{F} = q\mathbf{E} + q\mathbf{v} \times \mathbf{B}$$



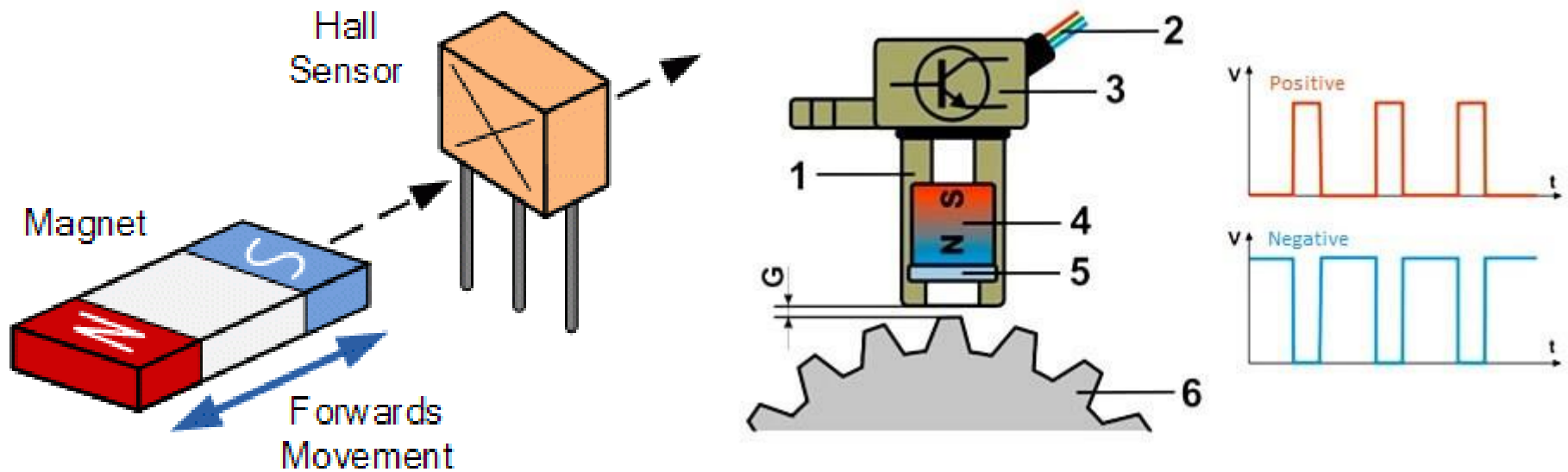
Magnetic (Hall) sensors



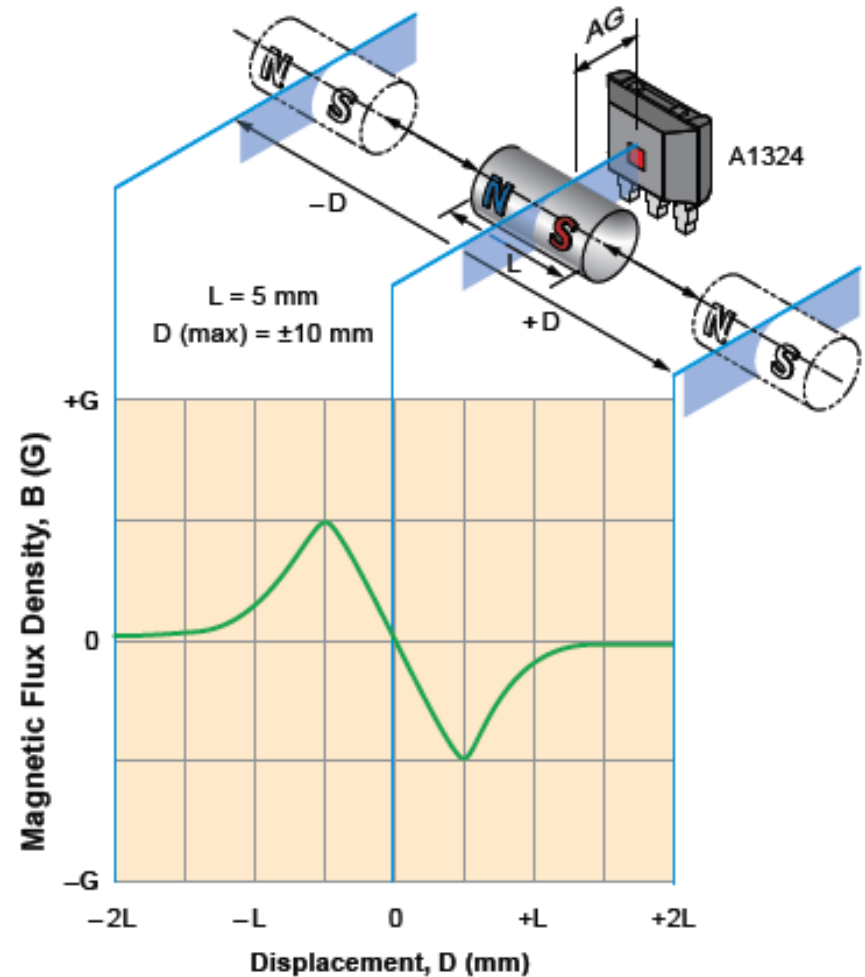
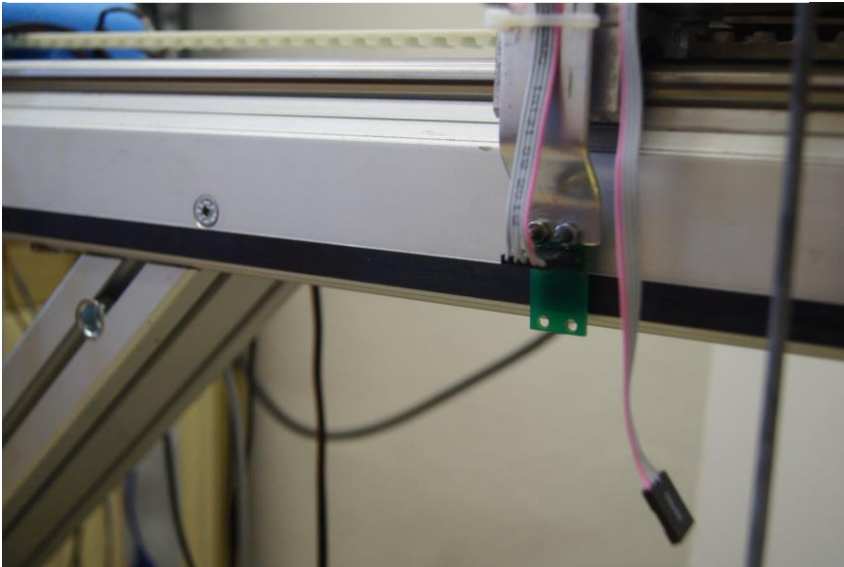
Magnetic (Hall) sensors



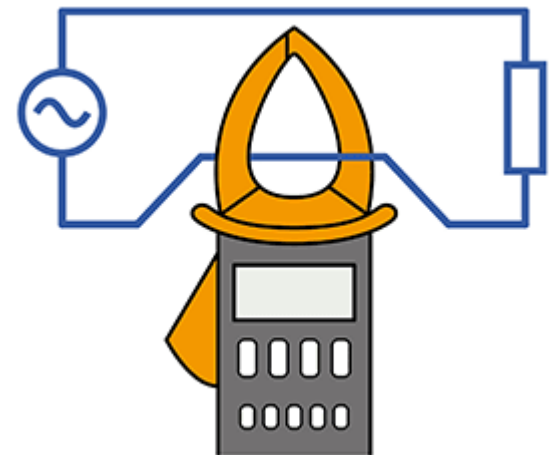
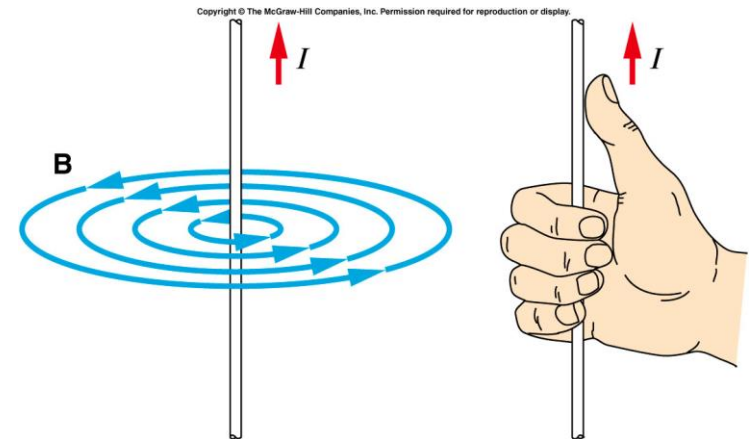
Magnetic (Hall) sensors



Magnetic (Hall) sensors



□ Clampmeter



Optical sensors

□ Time of flight (TOF) sensors

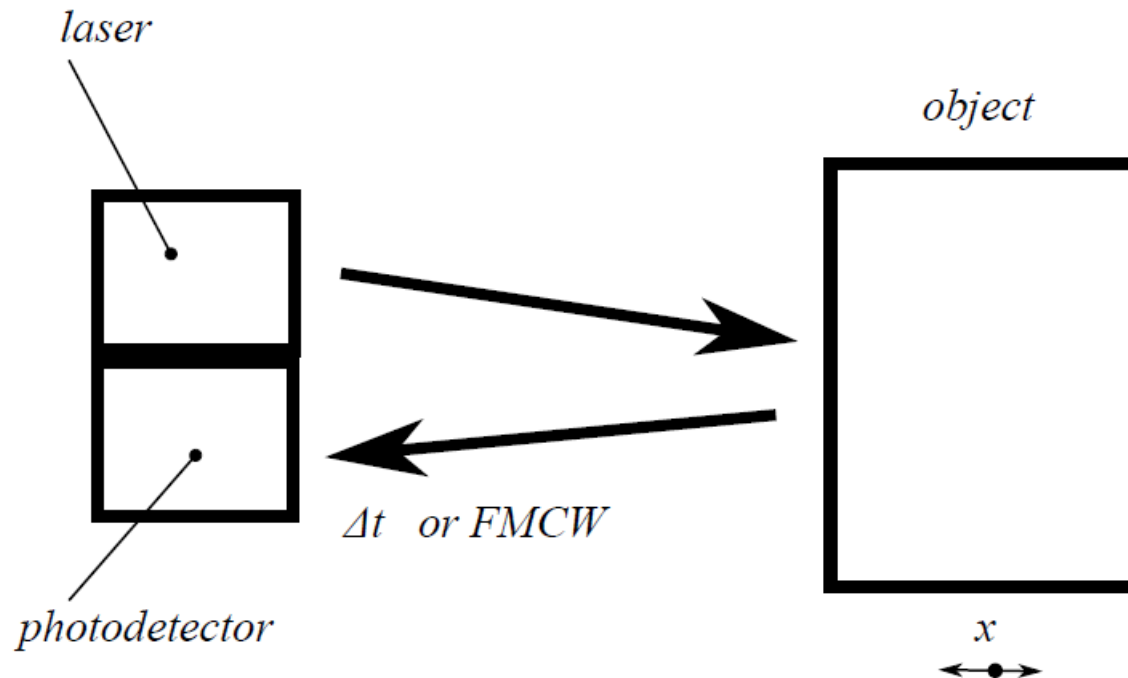
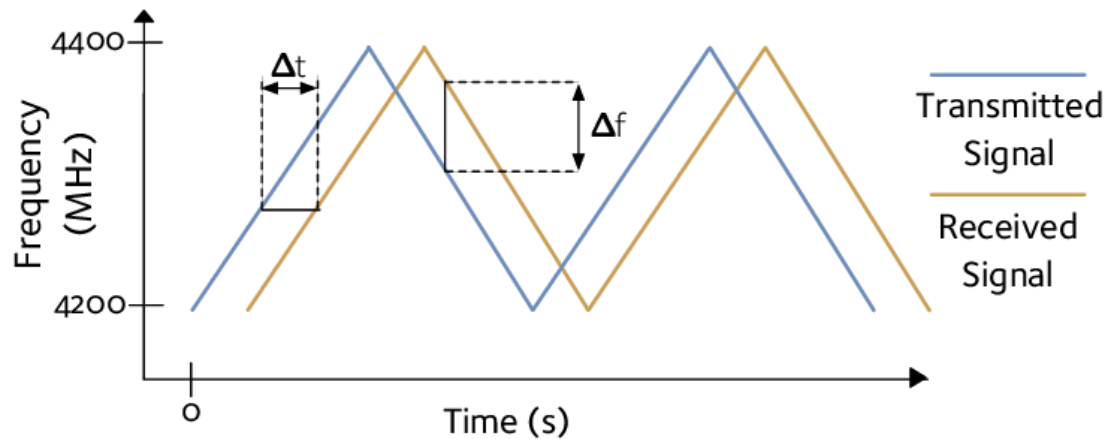


FIGURE 7.41: Optical distance sensor—TOF

Optical sensors

□ Frequency modulated continuous wave (FMCW)



Optical sensors

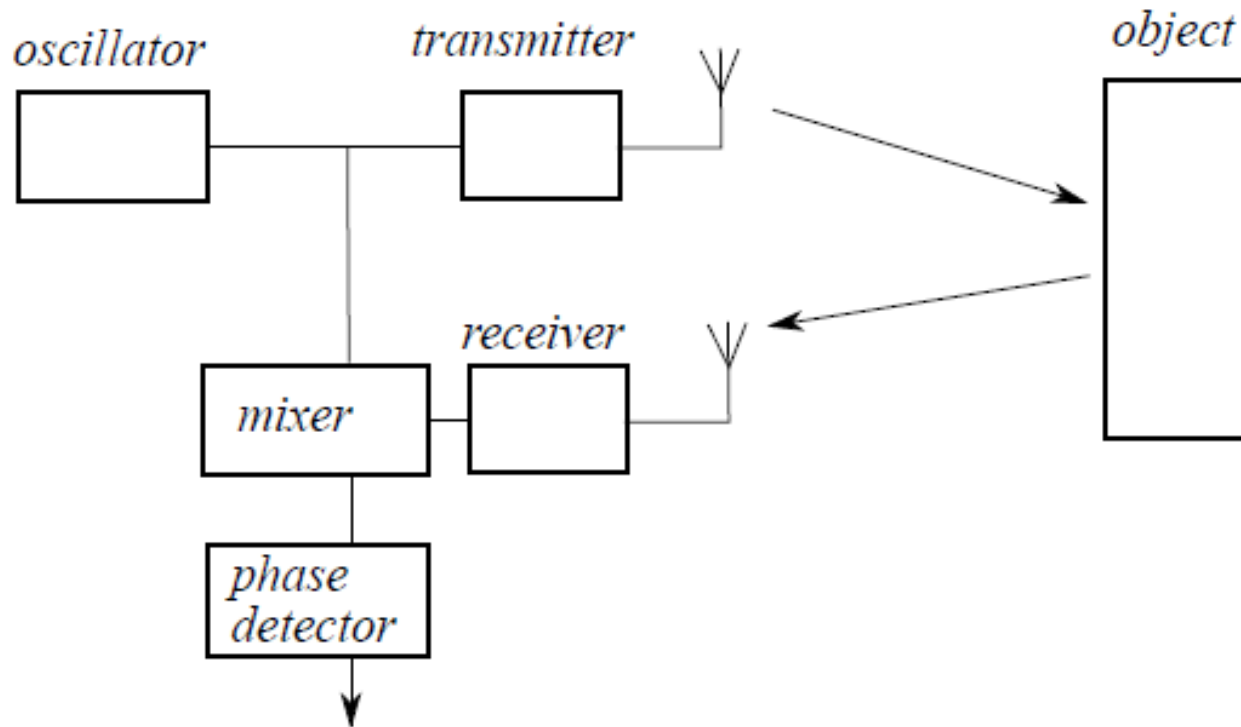
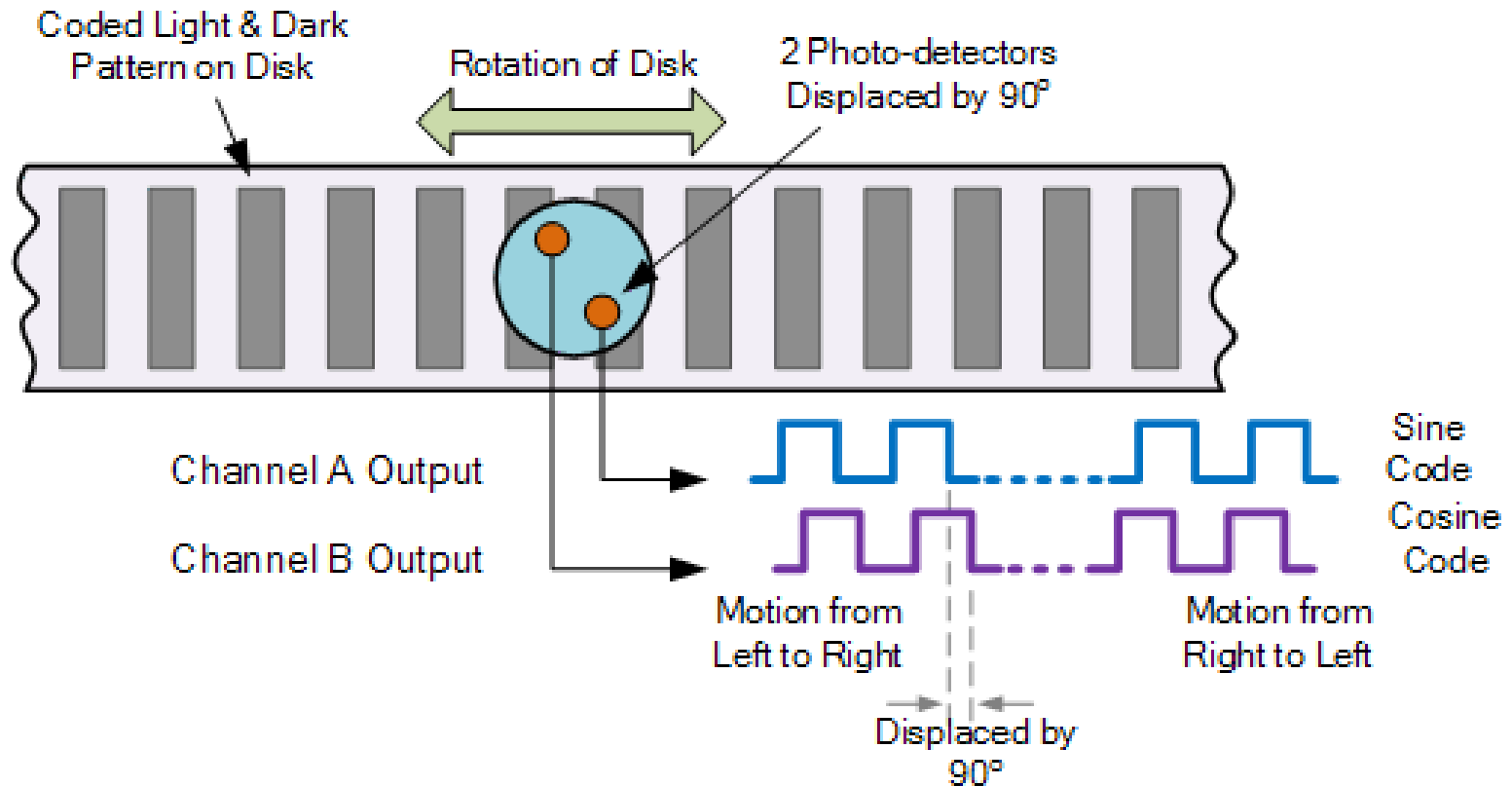
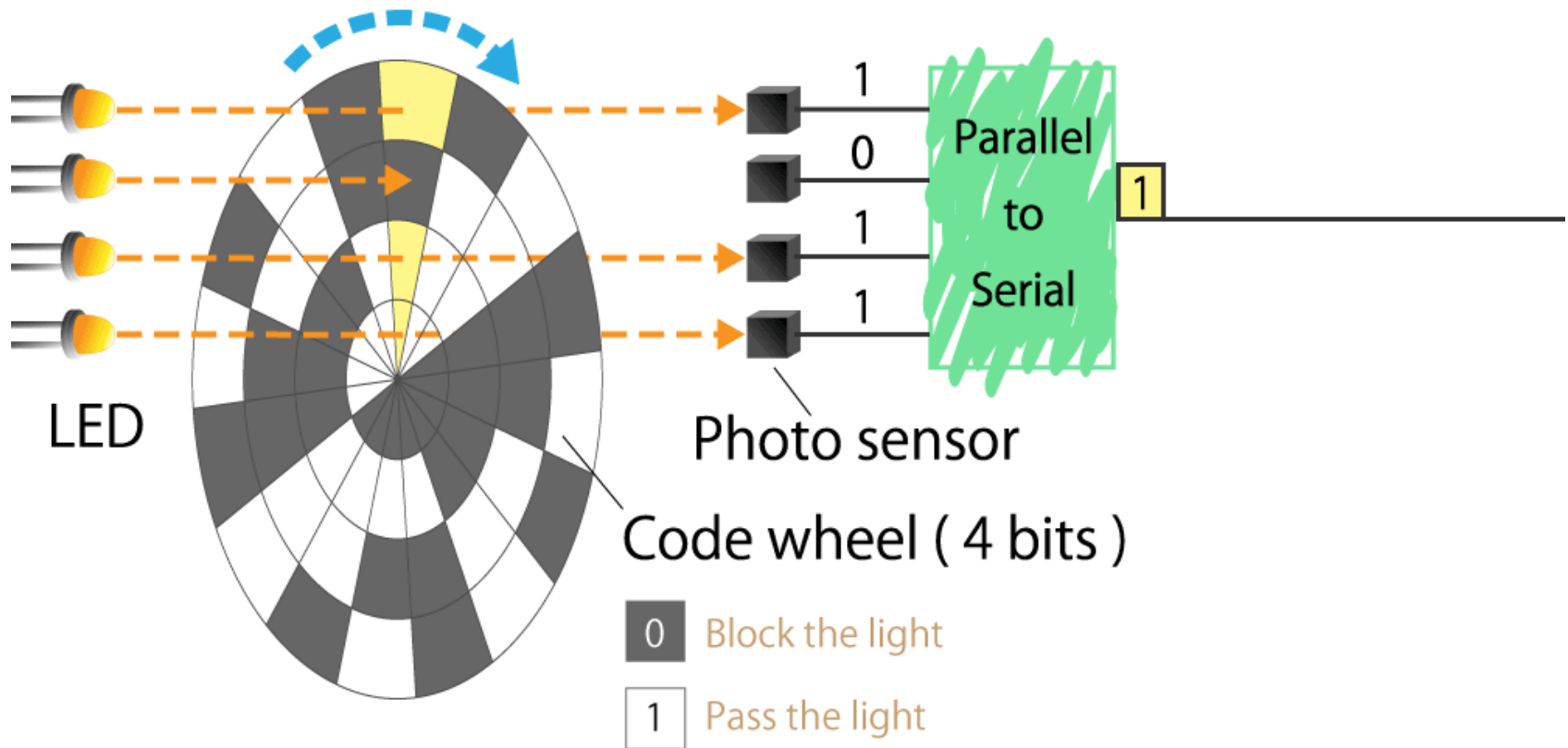


FIGURE 7.55: FMCW block diagram

Incremental rotary encoders (IRC)



Absolute rotary encoders



Inductive proximity sensors

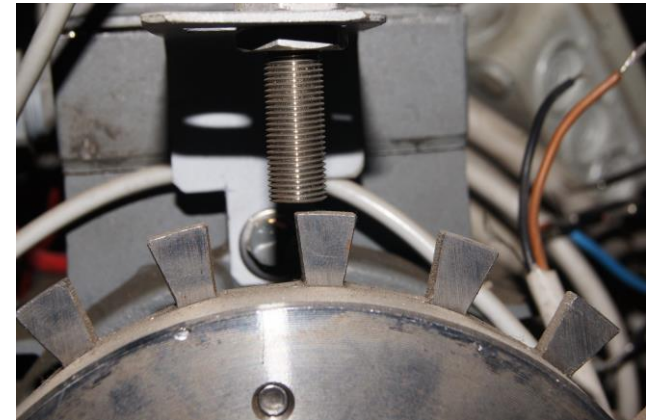
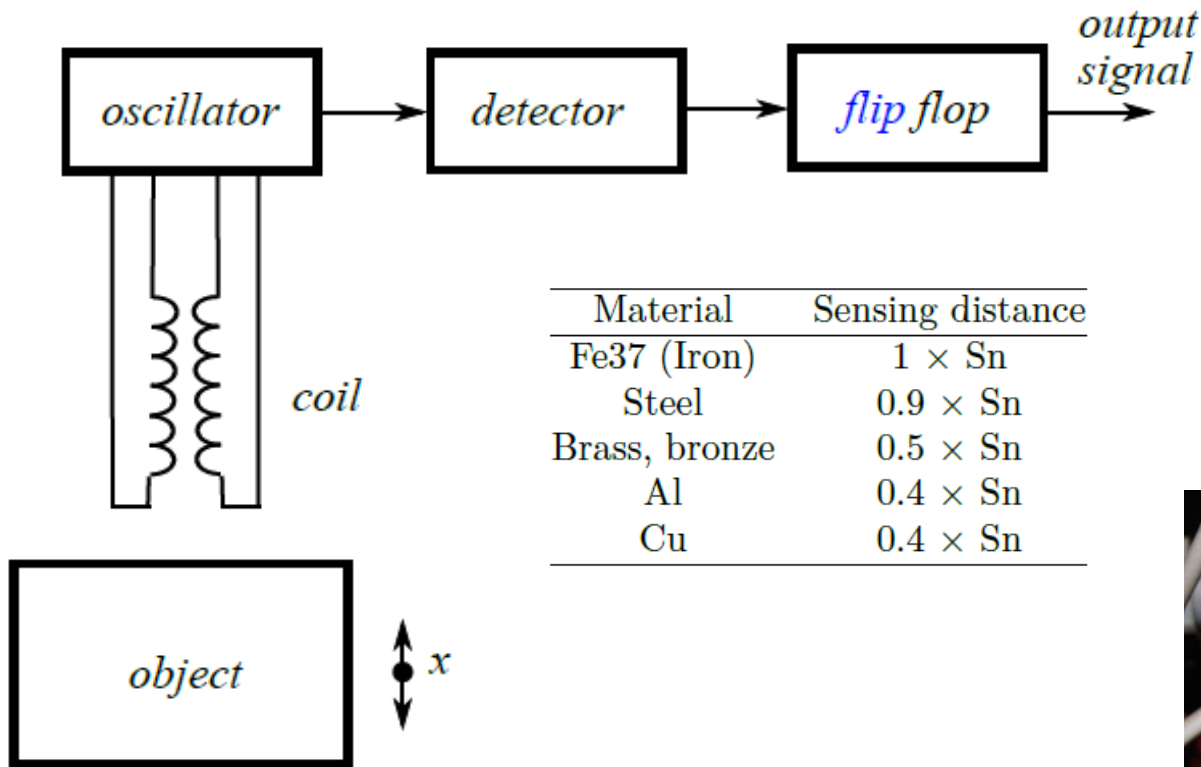


FIGURE 7.61: Principle of inductive proximity sensor

Only work with conductive objects

Capacitive proximity sensors

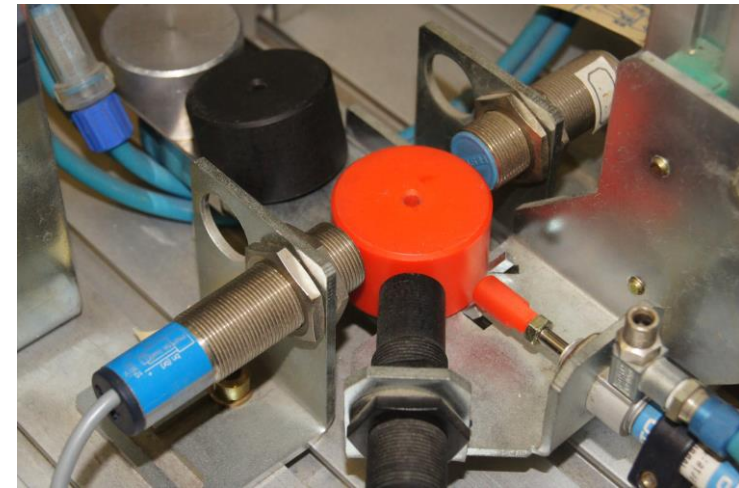
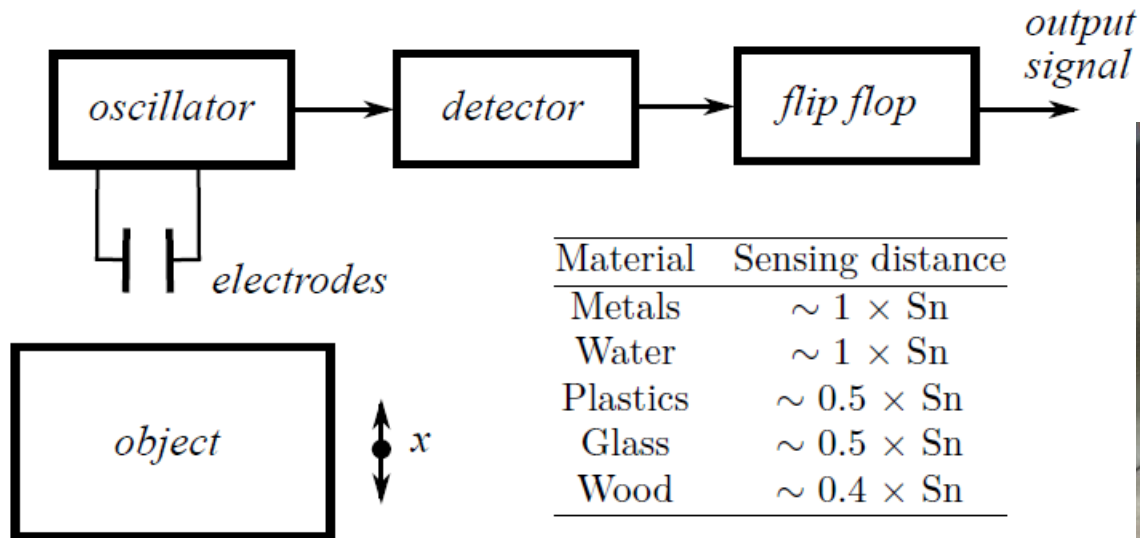


FIGURE 7.70: Principle of capacitive proximity sensor

Optic proximity sensors

